The Sharp Numerical-Range Spectral-Set Inequality

Qinyu Luo*
ByteDance, Building 1, Dazhongsi Square,
No. A18 North Third Ring Road West, Haidian District,
Beijing 100098, China

AI research collaborator: GPT-5.6 Sol Ultra (OpenAI)

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Abstract

We give a proof of the finite-dimensional scalar form of Crouzeix's conjecture: for every complex matrix A and every rational function r whose poles lie off the numerical range $W(A)$,

$$\|r(A)\|_2 \le 2 \max_{z\in W(A)} |r(z)|.$$

The argument first treats real-analytic strictly convex outer domains. It combines a positive normalized operator-valued double-layer kernel, conjugate Cauchy companions for all powers of finite Blaschke products, Fourier cancellation, and a bounded-sequence recurrence that excludes a singular value larger than 2. Schur approximation and a convex outer approximation then yield the polynomial and rational statements.

The theorem below is the sharp constant-2 assertion conjectured by Crouzeix [1]. Crouzeix later obtained the dimension-free bound 11.08 [2], and the previously published general bound is $1+\sqrt{2}$, due to Crouzeix and Palencia [5]. Two contemporaneous preprints also claim the sharp bound, by Jin [14] and by Lorist and Schwenninger [15]. The present proof is stated only for scalar-valued functions and finite-dimensional matrices.

Theorem 1. *Let $A \in \mathbb{C}^{n\times n}$, and let*

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n,\ \|x\|_2 = 1\}.$$

Then every complex polynomial p satisfies

$$\|p(A)\|_2 \le 2 \max_{z\in W(A)} |p(z)|.$$

The same inequality holds for every rational function whose poles lie off $W(A)$. Thus $W(A)$, which is compact in finite dimensions, is a 2-spectral set for A.

Proof. We first prove the inequality with $W(A)$ replaced by a particularly regular convex neighborhood. Let Ω be a bounded convex domain with real-analytic strictly convex boundary, and assume that $W(A) \Subset \Omega$. We shall prove

$$\|h(A)\|_2 \le 2\|h\|_{\infty,\Omega} \qquad (h \in H^\infty(\Omega)). \tag{1}$$

*Corresponding author: qinyuluo123@gmail.com

The conjugate-Cauchy and positive double-layer framework used below originates in Delyon and Delyon [3] and was developed for convex spectral sets and numerical-range functional calculus in [4, 2, 5]. For a clarification of the symmetrized mechanism and a recent systematic treatment of the double-layer potential, see [6, 7]. The finite-fiber kernel and the all-powers recurrence below are the forms used in the present argument.

We use counterclockwise orientation on $\partial\Omega$. At $\zeta \in \partial\Omega$, let $n(\zeta)$ be the outward unit normal, regarded as a complex number, and let ds denote arclength. Thus

$$d\zeta = in(\zeta)\, ds. \tag{2}$$

For a function u analytic on a neighborhood of $\overline{\Omega}$, define its conjugate Cauchy transform by

$$(\Gamma u)(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{u(\zeta)}}{\zeta - z}\, d\zeta, \qquad z \in \Omega. \tag{3}$$

We next establish the boundary kernel that drives the proof. Let $\phi : \Omega \to \mathbb{D}$ be a conformal map. Real-analyticity of the boundary and Schwarz reflection imply that ϕ extends conformally across $\partial\Omega$ [8, Ch. 3]. Let

$$f = B \circ \phi, \tag{4}$$

where B is a nonconstant finite Blaschke product. Thus f is analytic on a neighborhood of $\overline{\Omega}$ and $|f| = 1$ on $\partial\Omega$. If $\zeta = \zeta(s)$ is an arclength parametrization and $f(\zeta(s)) = e^{it(s)}$, then

$$\kappa(\zeta) := t'(s) > 0, \qquad f'(\zeta)n(\zeta) = f(\zeta)\kappa(\zeta). \tag{5}$$

Indeed, a finite Blaschke product restricts to an orientation-preserving covering of the unit circle. In particular, all boundary solutions of $f(\zeta) = \alpha$, $\alpha \in \mathbb{T}$, are simple.

For $\alpha \in \mathbb{T}$, put

$$K_\alpha(z) = \sum_{\substack{\zeta \in \partial\Omega \\ f(\zeta) = \alpha}} \frac{n(\zeta)}{\kappa(\zeta)(\zeta - z)}. \tag{6}$$

This kernel has positive real part at A. In fact, convexity gives

$$\operatorname{Re}\big(\overline{n(\zeta)}(\zeta I - A)\big) \succ 0,$$

because $W(A) \Subset \Omega$. If T is strictly accretive, then

$$\operatorname{Re}(T^{-1}) = T^{-*}(\operatorname{Re} T)T^{-1} \succeq 0. \tag{7}$$

Since

$$n(\zeta)(\zeta I - A)^{-1} = \big(\overline{n(\zeta)}(\zeta I - A)\big)^{-1},$$

we obtain

$$H_\alpha := \operatorname{Re} K_\alpha(A) \succeq 0. \tag{8}$$

Let $dm(\alpha) = dt/(2\pi)$ be normalized arclength measure on $\mathbb{T}$. The change of variables $\alpha = f(\zeta)$ and (5) give $dt = \kappa(\zeta)ds$ on every sheet of the boundary covering. Hence, for $m \geq 0$,

$$\begin{aligned} \int_{\mathbb{T}} \alpha^m K_\alpha(z)\, dm(\alpha) &= \frac{1}{2\pi} \int_{\partial\Omega} \frac{f(\zeta)^m n(\zeta)}{\zeta - z}\, ds \\ &= \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(\zeta)^m}{\zeta - z}\, d\zeta = f(z)^m, \end{aligned} \tag{9}$$

and, for $m \geq 1$,

$$\int_{\mathbb{T}} \overline{\alpha}^m K_\alpha(z)\, dm(\alpha) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{f(\zeta)^m}}{\zeta - z}\, d\zeta =: g_m(z), \tag{10}$$

where $g_m = \Gamma(f^m)$. These identities may be evaluated at A, since all functions involved are analytic on a neighborhood of $\operatorname{spec}(A)$. If $F = f(A)$ and $G_m = g_m(A)$, they yield

$$\int_{\mathbb{T}} H_\alpha\, dm(\alpha) = I, \qquad \int_{\mathbb{T}} \alpha^m H_\alpha\, dm(\alpha) = \frac{F^m + G_m^*}{2} \quad (m \geq 1). \tag{11}$$

The positivity in (8) and the first identity in (11) imply

$$\|F^m + G_m^*\|_2 \leq 2 \qquad (m \geq 1). \tag{12}$$

To see this directly, for unit vectors u, v use positivity first in the pointwise inequality $|v^* H_\alpha u|^2 \leq (v^* H_\alpha v)(u^* H_\alpha u)$, and then use Cauchy–Schwarz in $L^2(dm)$ to obtain

$$\left| v^* \left(\int_{\mathbb{T}} \alpha^m H_\alpha\, dm \right) u \right| \leq \left(\int_{\mathbb{T}} v^* H_\alpha v\, dm \right)^{1/2} \left(\int_{\mathbb{T}} u^* H_\alpha u\, dm \right)^{1/2} = 1.$$

The same scalar argument, with A replaced by any $z \in \Omega$, gives

$$|f(z)^m + \overline{g_m(z)}| \leq 2, \qquad \|g_m\|_{\infty,\Omega} \leq 3 \quad (m \geq 1). \tag{13}$$

We need one more consequence of the same kernel. Define

$$P_\alpha(z) = \frac{1}{1 - \overline{\alpha} f(z)}, \qquad R_\alpha(z) = P_\alpha(z) - K_\alpha(z), \qquad C_\alpha = \alpha I - F. \tag{14}$$

The principal part of P_α at a boundary solution $f(\zeta) = \alpha$ is

$$\frac{n(\zeta)}{\kappa(\zeta)(\zeta - z)},$$

by (5). Thus the apparent boundary poles cancel and R_α is analytic across $\partial\Omega$. Furthermore,

$$P_\alpha(z) = \sum_{m \geq 0} \overline{\alpha}^m f(z)^m \qquad (z \in \Omega).$$

Comparing its Fourier coefficients with (9)–(10) gives

$$\int_{\mathbb{T}} \alpha^m R_\alpha(z)\, dm(\alpha) = 0 \qquad (m \geq 0), \tag{15}$$

$$\int_{\mathbb{T}} \overline{\alpha}^m R_\alpha(z)\, dm(\alpha) = -g_m(z) \qquad (m \geq 1). \tag{16}$$

Only these Fourier-coefficient identities will be used; therefore no pointwise convergence of a Fourier series is required.

Since $I - \overline{\alpha} F = \overline{\alpha} C_\alpha$, we have $P_\alpha(A) = \alpha C_\alpha^{-1}$. Here C_α is invertible: the spectral mapping theorem and the maximum principle give $\operatorname{spec}(F) = f(\operatorname{spec}(A)) \subset \mathbb{D}$. A direct multiplication shows

$$C_\alpha^* \operatorname{Re}(P_\alpha(A)) C_\alpha = I - \operatorname{Re}(\overline{\alpha} F). \tag{17}$$

Using $P_\alpha = K_\alpha + R_\alpha$, we therefore obtain the exact decomposition

$$E_\alpha := I - \operatorname{Re}(\overline{\alpha}F) = Q_\alpha + C_\alpha^* \operatorname{Re}(R_\alpha(A))C_\alpha, \qquad Q_\alpha := C_\alpha^* H_\alpha C_\alpha \succeq 0. \tag{18}$$

We now prove that every f of the form (4) satisfies $\|f(A)\|_2 \le 2$. Suppose, to the contrary, that

$$\lambda := \|F\|_2 > 2. \tag{19}$$

Choose unit singular vectors x, y such that

$$Fx = \lambda y, \qquad F^*y = \lambda x. \tag{20}$$

For every function h analytic on Ω, put

$$\tau(h) = x^* h(A) x.$$

Because $h(A)$ commutes with F, (20) gives

$$y^* h(A) y = \tau(h), \qquad x^* h(A) y = \frac{1}{\lambda}\tau(fh). \tag{21}$$

Indeed,

$$y^* h(A) y = \lambda^{-2} x^* F^* h(A) F x = \lambda^{-2} x^* F^* F h(A) x = x^* h(A) x,$$

and the second equality follows from $y = Fx/\lambda$.

For $m \ge 1$, define the real numbers

$$\beta_m = \operatorname{Re}\tau(f^m g_m). \tag{22}$$

They are bounded uniformly in m. To make this completely explicit, choose a positively oriented contour $\gamma \Subset \Omega$ of winding number one around $\operatorname{spec}(A)$. The holomorphic functional calculus gives

$$\|h(A)\|_2 \le M \|h\|_{\infty,\Omega}, \qquad M := \frac{\operatorname{length}(\gamma)}{2\pi} \max_{z\in\gamma} \|(zI - A)^{-1}\|_2 < \infty. \tag{23}$$

Consequently, by (13),

$$|\beta_m| \le \|(f^m g_m)(A)\|_2 \le 3M \qquad (m \ge 1). \tag{24}$$

The case $m = 1$ of (12), together with (21), gives

$$\begin{aligned} 2 &\ge \operatorname{Re}\big(y^*(F + G_1^*)x\big) \\ &= \lambda + \frac{1}{\lambda}\operatorname{Re}\tau(fg_1) = \lambda + \frac{\beta_1}{\lambda}. \end{aligned} \tag{25}$$

Thus

$$\beta_1 \le -\lambda(\lambda - 2) < 1. \tag{26}$$

Fix an integer $N \ge 1$ and, for $\alpha \in \mathbb{T}$, set

$$v_N(\alpha) = x + \sum_{j=1}^{N} \overline{\alpha}^{\,j} F^{j-1} y. \tag{27}$$

The sum telescopes after multiplication by C_α:

$$w_N(\alpha) := C_\alpha v_N(\alpha) = \alpha x - (\lambda - 1)y - \overline{\alpha}^{\,N} F^N y. \tag{28}$$

Fourier orthogonality also gives the exact identity

$$\int_{\mathbb{T}} v_N(\alpha)^* E_\alpha v_N(\alpha)\, dm(\alpha) = 2 - \lambda. \tag{29}$$

For completeness, the mean of $\|v_N\|_2^2$ is

$$1 + \sum_{k=0}^{N-1} \|F^k y\|_2^2,$$

whereas the mean of $\overline{\alpha} v_N^* F v_N$ is

$$\lambda + \sum_{k=1}^{N-1} \|F^k y\|_2^2.$$

Their difference is precisely $2 - \lambda$.

Applying the positive decomposition (18) to v_N and using (29), we find

$$2 - \lambda \geq \int_{\mathbb{T}} w_N(\alpha)^* \operatorname{Re}(R_\alpha(A)) w_N(\alpha)\, dm(\alpha). \tag{30}$$

Put $d = \lambda - 1$. The three Fourier exponents occurring in w_N are $1, 0, -N$. Expanding the right-hand side and using (15)–(16) shows that the only nonzero pairings are

$$\int_{\mathbb{T}} w_N^* R_\alpha(A) w_N\, dm = d\, x^* G_1 y + x^* G_{N+1} F^N y - d\, y^* G_N F^N y. \tag{31}$$

By (21), the real part of (31) equals

$$\frac{\lambda - 1}{\lambda}\beta_1 + \frac{1}{\lambda}\beta_{N+1} - (\lambda - 1)\beta_N. \tag{32}$$

It follows from (30) that

$$\boxed{\beta_{N+1} \leq -\lambda(\lambda - 2) - (\lambda - 1)\beta_1 + \lambda(\lambda - 1)\beta_N} \qquad (N \geq 1). \tag{33}$$

This recurrence contradicts the uniform bound (24). Indeed, set

$$q = \lambda(\lambda - 1) > 1, \qquad c = -\lambda(\lambda - 2) - (\lambda - 1)\beta_1,$$

and define $U_1 = \beta_1$ and $U_{N+1} = c + qU_N$. Induction in (33) gives $\beta_N \leq U_N$. If

$$s = -\frac{c}{q - 1}$$

is the fixed point of this affine recurrence, then

$$(q - 1)(\beta_1 - s) = \lambda(\lambda - 2)(\beta_1 - 1) < 0 \tag{34}$$

by (19) and (26). Therefore

$$U_N = s + q^{N-1}(\beta_1 - s) \longrightarrow -\infty.$$

For large N this gives $\beta_N < -3M$, contradicting (24). The contradiction proves

$$\|(B \circ \phi)(A)\|_2 \leq 2 \tag{35}$$

for every finite Blaschke product B. The assertion is immediate for a constant unimodular B.

We now pass from finite Blaschke products to all bounded analytic functions. We record the elementary approximation argument. If $q : \mathbb{D} \to \overline{\mathbb{D}}$ is analytic, apply the Schur algorithm [9]

$$q_0 = q, \qquad \gamma_j = q_j(0), \qquad q_{j+1}(z) = \frac{q_j(z) - \gamma_j}{z(1 - \overline{\gamma_j} q_j(z))}.$$

As long as $|\gamma_j| < 1$, Schwarz's lemma shows that q_{j+1} is again a Schur function. At a chosen level N, replace q_N by a unimodular constant and reverse the recursion by

$$b_j(z) = \frac{\gamma_j + z b_{j+1}(z)}{1 + \overline{\gamma_j} z b_{j+1}(z)}. \tag{36}$$

The resulting b_0 is a finite Blaschke product. Each reverse step in (36) raises by one the order of vanishing of the difference from the original Schur iterate. Hence $b_0 - q = O(z^N)$. Since $|b_0 - q| \le 2$, the generalized Schwarz lemma even gives

$$|b_0(z) - q(z)| \le 2|z|^N.$$

If a Schur parameter is unimodular, the algorithm has already terminated in a finite Blaschke product. We have thus proved that finite Blaschke products are locally uniformly dense in the Schur class.

Given $h \in H^\infty(\Omega)$ with $\|h\|_{\infty,\Omega} \le 1$, apply this approximation to $q = h \circ \phi^{-1}$. If $B_j \to q$ locally uniformly on $\mathbb{D}$, then $B_j \circ \phi \to h$ locally uniformly on Ω. A Cauchy integral over a fixed contour surrounding $\operatorname{spec}(A)$ gives

$$(B_j \circ \phi)(A) \longrightarrow h(A).$$

Passing to the limit in (35) proves (1).

It remains to let the regular convex domains approach $W(A)$. By the Toeplitz–Hausdorff theorem [10, 11], $W(A)$ is convex. The standard compression proof recalls that, in dimension at most two, the numerical range is a (possibly degenerate) elliptical disk [12]; it therefore contains the segment joining any two prescribed points of $W(A)$. Compactness follows from continuity on the unit sphere. We briefly justify the needed geometric approximation. For the support-function and curvature-measure facts used below, see [13, Chs. 1 and 4]. For any compact convex set $K \subset \mathbb{C}$, its support function

$$h_K(\theta) = \max_{z \in K} \operatorname{Re}(e^{-i\theta} z)$$

is continuous and the distribution $h_K + h_K''$ is a nonnegative measure. Convolve h_K with a periodic Poisson kernel, add its uniform approximation error, and then add a positive constant tending to zero. The resulting real-analytic support functions h_j satisfy

$$h_j > h_K, \qquad h_j + h_j'' > 0, \qquad h_j \longrightarrow h_K \quad \text{uniformly.}$$

The curves

$$\zeta_j(\theta) = e^{i\theta}\big(h_j(\theta) + i h_j'(\theta)\big)$$

bound real-analytic strictly convex domains Ω_j with $K \Subset \Omega_j$, and $\overline{\Omega}_j \to K$ in Hausdorff distance.

Apply this construction to $K = W(A)$. From (1),

$$\|p(A)\|_2 \le 2 \max_{z \in \overline{\Omega}_j} |p(z)|.$$

Because p is continuous and $\overline{\Omega}_j \to W(A)$ in Hausdorff distance, the right-hand side tends to $2\max_{z\in W(A)}|p(z)|$. This proves the theorem.

Finally, if r is rational and has no pole on $W(A)$, choose the domains Ω_j above sufficiently close to $W(A)$ that they contain no pole of r. Applying (1) directly to r on each Ω_j and then taking the same Hausdorff limit gives

$$\|r(A)\|_2 \le 2 \max_{z\in W(A)} |r(z)|.$$

This proves the stated spectral-set formulation as well. □

Sharpness. The constant cannot be reduced. For

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad p(z) = z,$$

the numerical range is the closed disk of radius $1/2$ centered at the origin, whereas $\|p(A)\|_2 = \|A\|_2 = 1$. Thus the ratio in the theorem is exactly 2.

Author Contributions

Qinyu Luo: conceptualization, methodology, formal analysis, writing—original draft, writing—review and editing, validation, supervision, and project administration. The author approved the final manuscript and accepts full responsibility for all of its contents.

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GPT-5.6 Sol Ultra (OpenAI), accessed through Codex, served as an AI research collaborator. It made substantial non-author contributions through adversarial line-by-line proof auditing; algebraic, Fourier-sign, boundary, and extremal-case checks; design and execution of three-route numerical cross-checks; literature and provenance verification; and English-language editing and release preparation. In accordance with current publication ethics rules, the AI system is not entered in the formal author metadata: it cannot approve the submitted version, hold copyright, or assume legal and scholarly accountability. Qinyu Luo reviewed the outputs, made the final decisions, and assumes complete responsibility for the manuscript.

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